

Mutual Correlated Agreement for Reed–Solomon Codes up to Capacity in the Low-Rate Regime*

Abstract

We prove near-capacity mutual correlated agreement for Reed–Solomon codes over arbitrary evaluation sets in the low-rate regime. Fix any constant $\theta \in (0, 1)$ and sufficiently small constant $\varepsilon > 0$. For rate $(1 - \theta)\varepsilon$ and agreement εn , every affine line of received words has only $n^{O_{\theta, \varepsilon}(1)}$ bad parameters. Consequently, the mutual-correlated-agreement error at radius $1 - \varepsilon$ is at most $n^{O_{\theta, \varepsilon}(1)}/q$.

1 Introduction

Reed–Solomon (RS) codes are among the most fundamental error-correcting codes and have found numerous applications throughout theoretical computer science and cryptography. In several of these applications, including polynomial commitment schemes and proximity testing, ordinary list decoding is not quite enough. Mutual correlated agreement (MCA) asks a related question for a structured family of received words: can many different members of the family agree well with the code for unrelated reasons? [Arn+24, Hab25]

The simplest family is an affine line

$$u_0 + zu_1, \quad z \in \mathbb{F}_q.$$

Suppose that for some z , a low-degree polynomial agrees with $u_0 + zu_1$ on many evaluation points. We would like that agreement to come from two fixed low-degree polynomials, one matching u_0 and one matching u_1 on the same coordinates. A value of z for which this fails is a *bad parameter*.

For arbitrary RS evaluation domains, MCA is known through the Johnson radius by adapting the Guruswami–Sudan interpolation argument [Hab25, Ben+25]. General reductions from list decoding to MCA are also known, but they incur an additional Johnson-type square-root loss [Gao+25]. Near-capacity results are known for several structured or random code families [GG26, Bor+26]. For worst-case RS evaluation domains, Jo [Jo26] recently gave the first deterministic result beyond the Johnson radius, by a fixed number of integer error positions.

Theorem 4.1 of the recent list-decoding paper of Brakensiek, Chen, Putterman, Zhang, and Zheng [BCPZZ26] gives a much stronger result for arbitrary RS domains: for fixed $\theta \in (0, 1)$ and sufficiently small $\varepsilon > 0$, codes of rate

$$R \leq (1 - \theta)\varepsilon$$

can be efficiently list decoded from a $1 - \varepsilon$ fraction of errors. Their main interpolation result is Proposition 3.13. It produces a polynomial

$$Q(X, Y_0, Y_1, \dots, Y_d)$$

whose low-degree polynomial solutions contain every sufficiently agreeing codeword. We use that interpolation result as our main input.

*This paper was AI generated.

Our goal here is to show that the same agreement threshold still works when the received word moves along an affine line. Most of the interpolation machinery is inherited from [BCPZZ26]; the new part is controlling how the polynomial solutions of the resulting differential equation vary with the line parameter.

Our result. We obtain the same near-capacity radius as the list decoder, with polynomially many bad line parameters.

Theorem 1.1 (Main Result, Informal Version). *Fix $\theta \in (0, 1)$. There exists $\varepsilon_0(\theta) > 0$ such that for every fixed $0 < \varepsilon < \varepsilon_0(\theta)$ there is $C_{\theta, \varepsilon} > 0$ for which the following holds for all sufficiently large n . Let*

$$k = \lfloor (1 - \theta)\varepsilon n \rfloor,$$

let $q \geq n$ be prime, and let $\alpha_1, \dots, \alpha_n \in \mathbb{F}_q$ be distinct. Then for every $u_0, u_1 \in \mathbb{F}_q^n$,

$$|\text{Bad}_{\lceil \varepsilon n \rceil}(u_0, u_1)| \leq n^{C_{\theta, \varepsilon}}.$$

Moreover,

$$\varepsilon_{\text{MCA}}(\text{RS}_{<k}(\alpha_1, \dots, \alpha_n), \lceil \varepsilon n \rceil) \leq \min \left\{ 1, \frac{n^{C_{\theta, \varepsilon}}}{q} \right\}.$$

Thus, over a sufficiently large polynomial-size field, the MCA error is inverse polynomial. More importantly, the agreement threshold is exactly the one achieved by the list decoder in [BCPZZ26]; there is no additional Johnson-type loss.

Proof overview. The proof has three steps. First, we take the interpolation system from Section 3 of [BCPZZ26] and replace each received symbol by

$$u_{0,i} + Zu_{1,i}.$$

The proof of Proposition 3.13 has constant-factor dimension slack, so after shrinking $\varepsilon_0(\theta)$ if needed, the same linear-algebra argument works over $\mathbb{F}_q[Z]$. We obtain a single interpolation polynomial of constant Z -degree whose specialization is nonzero for every $z \in \mathbb{F}_q$. Hence every polynomial with at least εn agreements with $u_0 + zu_1$ satisfies one specialized differential equation.

Second, we study all polynomial solutions of this equation at once as z varies. We follow Kopparty's differential-equation solver [Kop15]. At a regular solution, the polynomial is determined by z , one expansion point, and a bounded number of initial Hasse derivatives. Kopparty's lifting recurrence then expresses all later derivatives, and hence the polynomial itself, as polynomial functions of this initial data. If the separant vanishes identically, we recurse on a lower-degree differential equation. This produces only $O_{\theta, \varepsilon}(1)$ solution families.

Finally, on each solution family we impose the agreement equations

$$P(\alpha_i) = u_{0,i} + zu_{1,i}.$$

If k of these equations vanish identically on one irreducible family, interpolation forces that family onto one affine codeword line, which contributes at most n bad parameters. Otherwise, every bad point lies on at least $\lceil \varepsilon n \rceil - k + 1 = \Omega_{\theta, \varepsilon}(n)$ equations that properly cut the family. An induction on dimension, together with Bézout, then gives a polynomial bound.

Organization. Section 2 records the notation and algebraic facts we use. Section 3 interpolates along an affine line. Section 4 describes the polynomial solution families. Section 5 counts bad parameters and proves the main theorem.

2 Preliminaries

Unless otherwise stated, $q \geq n$ is prime and $\alpha_1, \dots, \alpha_n \in \mathbb{F}_q$ are distinct.

Hasse derivatives. For $P \in \mathbb{F}_q[X]$, define its j th Hasse derivative by

$$P(X + T) = \sum_{j \geq 0} P^{[j]}(X) T^j. \quad (1)$$

In particular,

$$P(\alpha + T) = \sum_{j \geq 0} P^{[j]}(\alpha) T^j, \quad (2)$$

and applying (1) to $P^{[i]}$ gives

$$P^{[i]}(\alpha + T) = \sum_{t \geq 0} \binom{i+t}{i} P^{[i+t]}(\alpha) T^t. \quad (3)$$

We refer to an identity

$$G(z, X, P, P^{[1]}, \dots, P^{[r]}) \equiv 0$$

as a polynomial differential equation. If r is the largest derivative index on which G depends, we call

$$\frac{\partial G}{\partial Y_r}$$

the *separant* of G .

We use the following lifting theorem of Kopparty [Kop15, Theorem 4.4], stated in Hasse-derivative notation.

Lemma 2.1 (Regular lifting). *Let $\mathbb{F}$ have characteristic p , let $G(X, Y_0, \dots, Y_r) \in \mathbb{F}[X, Y_0, \dots, Y_r]$, and let $t \geq 1$ satisfy*

$$\binom{t+r}{r} \neq 0 \quad \text{in } \mathbb{F}.$$

If $f \in \mathbb{F}[X]$ and $\alpha \in \mathbb{F}$ satisfy

$$\frac{\partial G}{\partial Y_r}(\alpha, f(\alpha), f^{[1]}(\alpha), \dots, f^{[r]}(\alpha)) \neq 0$$

and

$$G(X, f, f^{[1]}, \dots, f^{[r]}) \equiv 0 \pmod{(X - \alpha)^t},$$

then there is a unique $\gamma \in \mathbb{F}$ such that, for

$$g(X) = f(X) + \gamma(X - \alpha)^{t+r},$$

we have

$$G(X, g, g^{[1]}, \dots, g^{[r]}) \equiv 0 \pmod{(X - \alpha)^{t+1}}.$$

The proof of [Kop15, Theorem 4.4] gives the update explicitly. If c_t is the coefficient of $(X - \alpha)^t$ in $G(X, f, f^{[1]}, \dots, f^{[r]})$, then

$$\gamma = -\frac{c_t}{\binom{t+r}{r} \frac{\partial G}{\partial Y_r}(\alpha, f(\alpha), \dots, f^{[r]}(\alpha))}. \quad (4)$$

Hence, when $q > k + r$, the successive Taylor coefficients of a degree- $< k$ regular solution are determined by its first $r + 1$ Hasse derivatives at one point. This is the regular case of Kopparty's algorithm SOLVE; see also [Kop15, Corollary 4.5 and Section 4.2].

Algebraic sets. For polynomials $F_1, \dots, F_t$, write

$$\mathcal{Z}(F_1, \dots, F_t) = \{x : F_1(x) = \dots = F_t(x) = 0\}.$$

All algebraic sets are over $\mathbb{F}_q$. We use $\dim V$, $\deg V$, and $\text{Irr}(V)$ in the usual sense. We only need two standard consequences of Bézout [Har92]: if V is irreducible and F does not vanish identically on V , then every component of $V \cap \mathcal{Z}(F)$ has smaller dimension and

$$\sum_{W \in \text{Irr}(V \cap \mathcal{Z}(F))} \deg W \leq \deg(V) \deg(F). \quad (5)$$

Also, a zero-dimensional algebraic set has at most its degree many geometric points.

Reed–Solomon codes and mutual correlated agreement. Define

$$\text{RS}_{<k}(\alpha_1, \dots, \alpha_n) = \{(P(\alpha_1), \dots, P(\alpha_n)) : P \in \mathbb{F}_q[X], \deg P < k\}.$$

For $P \in \mathbb{F}_q[X]$ and $y \in \mathbb{F}_q^n$, write

$$\text{agr}(P, y) = |\{i \in [n] : P(\alpha_i) = y_i\}|.$$

Throughout the rest of the paper, fix $\theta \in (0, 1)$ and sufficiently small $\varepsilon > 0$, and set

$$A = \lceil \varepsilon n \rceil, \quad k = \lfloor (1 - \theta)\varepsilon n \rfloor, \quad d = \lceil \varepsilon^{-3/\theta} \rceil. \quad (6)$$

Fix $u_0, u_1 \in \mathbb{F}_q^n$. We say that $z \in \mathbb{F}_q$ is *bad at agreement A* if there are $I \subseteq [n]$, $|I| \geq A$, and $P \in \mathbb{F}_q[X]_{<k}$ such that

$$P(\alpha_i) = u_{0,i} + zu_{1,i} \quad (i \in I),$$

but there do not exist $P_0, P_1 \in \mathbb{F}_q[X]_{<k}$ satisfying

$$P_j(\alpha_i) = u_{j,i} \quad (j \in \{0, 1\}, i \in I).$$

Write $\text{Bad}_A(u_0, u_1)$ for the set of bad parameters. Since $A > k$ for all sufficiently large n , if such P_0, P_1 exist then the witness polynomial also satisfies $P = P_0 + zP_1$ identically.

Finally, define

$$\varepsilon_{\text{MCA}}(\text{RS}_{<k}(\alpha_1, \dots, \alpha_n), A) = \max_{u_0, u_1 \in \mathbb{F}_q^n} \frac{|\text{Bad}_A(u_0, u_1)|}{q}.$$

Interpolation from the list-decoding theorem. We use the interpolation space $\mathcal{Q}$ from Section 3 of [BCPZZ26]. The following is exactly the part of Proposition 3.13 that we need.

Proposition 2.2 ([BCPZZ26], Proposition 3.13). *For every $y \in \mathbb{F}_q^n$, there is a nonzero $Q \in \mathcal{Q}$ such that*

$$\text{agr}(P, y) \geq A \implies Q(X, P, P^{[1]}, \dots, P^{[d]}) \equiv 0$$

for every $P \in \mathbb{F}_q[X]_{<k}$. Moreover,

$$\deg_X Q = O_{\theta, \varepsilon}(n), \quad \deg_{Y_i} Q = O_{\theta, \varepsilon}(1) \quad (0 \leq i \leq d).$$

3 Interpolation Along a Formal Line

Fix $u_0, u_1 \in \mathbb{F}_q^n$. We first find one interpolation polynomial that works simultaneously for every received word on the affine line.

Proposition 3.1. *There is a nonzero polynomial*

$$\widehat{Q}(Z, X, Y_0, \dots, Y_d) \in \mathbb{F}_q[Z, X, Y_0, \dots, Y_d]$$

such that

$$\deg_Z \widehat{Q}, \deg_{Y_i} \widehat{Q} = O_{\theta, \varepsilon}(1), \quad \deg_X \widehat{Q} = O_{\theta, \varepsilon}(n),$$

$$\widehat{Q}(z, X, Y_0, \dots, Y_d) \neq 0 \quad (z \in \mathbb{F}_q),$$

and

$$\text{agr}(P, u_0 + zu_1) \geq A \implies \widehat{Q}(z, X, P, P^{[1]}, \dots, P^{[d]}) \equiv 0$$

for every $z \in \mathbb{F}_q$ and $P \in \mathbb{F}_q[X]_{<k}$.

Proof. Let $b = O_{\theta, \varepsilon}(1)$ bound the Y_0 -degree of Q , and let r be the one-point local rank bound from Lemma 3.12 of [BCPZZ26]. The parameter calculation in the proof of Proposition 3.13 of [BCPZZ26] has constant-factor slack; after decreasing $\varepsilon_0(\theta)$ if necessary, we may assume

$$\dim Q > 2nr. \tag{7}$$

Consider polynomials

$$\sum_{j=0}^b Z^j Q_j(X, Y_0, \dots, Y_d), \quad Q_j \in Q.$$

There are $(b+1)\dim Q$ coefficients to choose. At coordinate i , replace the received symbol in the local constraints from Section 3 of [BCPZZ26] by $u_{0,i} + Zu_{1,i}$. The chosen polynomial has Z -degree at most b , and the substitution contributes at most another b because the Y_0 -degree is at most b . Thus every local expression has Z -degree at most $2b$.

For each fixed coefficient of Z , the same translated-and-truncated map from [BCPZZ26] lands in a space of dimension at most r . Hence one coordinate contributes rank at most $(2b+1)r$, and all n coordinates contribute rank at most

$$n(2b+1)r < 2n(b+1)r < (b+1)\dim Q$$

by (7). Therefore the homogeneous system has a nonzero solution $\widehat{Q}$.

Now divide the coefficient polynomials of $\widehat{Q}$ in $\mathbb{F}_q[Z]$ by their common gcd. The local constraints are homogeneous, so this preserves them. After this normalization, no specialization $\widehat{Q}(z, \cdot)$ can be the zero polynomial: if it were, every coefficient would be divisible by $Z - z$, contradicting gcd 1.

Finally, fix $z \in \mathbb{F}_q$. The polynomial $\widehat{Q}(z, \cdot)$ has the same monomial support and satisfies the same local constraints as the interpolation polynomial in [BCPZZ26]. The contact and degree-versus-multiplicity argument at the end of the proof of Proposition 3.13 therefore applies unchanged. Hence every P with at least A agreements satisfies

$$\widehat{Q}(z, X, P, P^{[1]}, \dots, P^{[d]}) \equiv 0.$$

The degree bounds follow from the construction and Proposition 2.2. $\square$

4 Polynomial Solution Families

Fix $\widehat{Q}$ from Proposition 3.1. We now study the degree- $< k$ polynomial solutions of

$$\widehat{Q}(z, X, P, P^{[1]}, \dots, P^{[d]}) \equiv 0 \quad (8)$$

as z varies. We follow Kopparty's algorithm SOLVE [Kop15, Section 4.2], while keeping the line parameter and the initial derivative values as variables.

4.1 Regular Solutions

Consider a nonzero polynomial

$$G(Z, X, Y_0, \dots, Y_r)$$

that occurs in the recursion, where r is the largest derivative index on which G depends. Set

$$S = \frac{\partial G}{\partial Y_r}.$$

Introduce one more variable U and define

$$B_G = \mathcal{Z}(G(z, \alpha, \beta_0, \dots, \beta_r), US(z, \alpha, \beta_0, \dots, \beta_r) - 1). \quad (9)$$

Lemma 4.1. *Regular lifting defines a polynomial $P_G(X)$ of degree $< k$ on B_G whose coefficients have degree $O_{\theta, \varepsilon}(n^2)$.*

Proof. By Lemma 2.1, the Taylor coefficients are determined recursively. We only need to track their degrees.

Let $\beta_j = P^{[j]}(\alpha)$. By (3),

$$P^{[i]}(\alpha + T) = \sum_{t \geq 0} \binom{i+t}{i} \beta_{i+t} T^t.$$

Substitute these expansions and $X = \alpha + T$ into $G(z, X, P, P^{[1]}, \dots, P^{[r]})$. For $t \geq 1$, the new coefficient β_{r+t} can only enter the coefficient of T^t through the linear contribution of $P^{[r]}$ to one occurrence of Y_r . Thus that coefficient has the form

$$\binom{r+t}{r} S \beta_{r+t} + R_t, \quad (10)$$

where R_t depends only on $\beta_0, \dots, \beta_{r+t-1}$. Since $r+t < k+d < q$, the binomial coefficient is nonzero. On B_G , $US = 1$, so

$$\beta_{r+t} = -\binom{r+t}{r}^{-1} U R_t. \quad (11)$$

To bound degrees, suppose a monomial of R_t contains β_{r+j} with multiplicity a_j . Such a factor can only enter with T -degree at least j , so every monomial contributing to the coefficient of T^t satisfies

$$\sum_{j=1}^{t-1} j a_j \leq t. \quad (12)$$

Assume inductively that β_{r+j} has degree at most $C_{\theta, \varepsilon} n j$ for $j < t$. The contribution coming directly from $G(z, \alpha + T, \dots)$ has degree $O_{\theta, \varepsilon}(n)$ in the initial variables, while (12) bounds the extra degree from previously determined coefficients by $C_{\theta, \varepsilon} n t$. Equation (11) therefore preserves an $O_{\theta, \varepsilon}(n t)$ bound. Since $t < k = O_{\theta, \varepsilon}(n)$, every Taylor coefficient has degree $O_{\theta, \varepsilon}(n^2)$.

Finally set

$$P_G(X) = \sum_{j=0}^{k-1} \beta_j (X - \alpha)^j.$$

Changing from the Taylor basis at α to the monomial basis in X introduces only powers of α of degree at most k , so the coefficient degrees remain $O_{\theta, \varepsilon}(n^2)$. $\square$

4.2 Covering All Polynomial Solutions

We now iterate the regular and singular cases of Kopparty's recursion.

Proposition 4.2. *There are a set $E \subseteq \mathbb{F}_q$ of size $O_{\theta, \varepsilon}(1)$ and $O_{\theta, \varepsilon}(1)$ algebraic sets B , each equipped with a polynomial $P(X)$ of degree $< k$, such that every solution of (8) with $z \notin E$ is represented by one of these sets. Moreover,*

$$\dim B \leq d + 2,$$

the coefficients of every P have degree $O_{\theta, \varepsilon}(n^2)$, and the sum of the degrees of all irreducible components of all the sets B is $O_{\theta, \varepsilon}(n^2)$.

Proof. Run Kopparty's recursion [Kop15, Section 4.2]. At a recursion polynomial G , let $S = \partial G / \partial Y_r$.

For the regular branch, use B_G from (9). Projection away from U identifies B_G with the locus

$$G(z, \alpha, \beta_0, \dots, \beta_r) = 0, \quad S(z, \alpha, \beta_0, \dots, \beta_r) \neq 0.$$

There are $r + 3$ ambient variables and one nonzero equation, so $\dim B_G \leq r + 2 \leq d + 2$. The two defining equations $G = 0$ and $US - 1 = 0$ have degree $O_{\theta, \varepsilon}(n)$, hence Bézout gives

$$\sum_{V \in \text{Irr}(B_G)} \deg V = O_{\theta, \varepsilon}(n^2).$$

Lemma 4.1 gives the corresponding polynomial family P_G .

These regular sets cover every regular solution. Indeed, if the separant does not vanish identically on a solution P , choose an expansion point α where it is nonzero, set $\beta_i = P^{[i]}(\alpha)$ and $U = S^{-1}$, and apply regular lifting.

If instead the separant vanishes identically, then P also solves $S = 0$, so we recurse on S . Differentiating with respect to Y_r strictly lowers the Y_r -degree until Y_r disappears; then the largest derivative index drops. Since all derivative-variable degrees are $O_{\theta, \varepsilon}(1)$, this process has only $O_{\theta, \varepsilon}(1)$ stages.

There is one small specialization issue in Z . At each stage we divide the coefficient polynomials by their gcd in $\mathbb{F}_q[Z]$ and put the roots of that gcd into an exceptional set E . The Z -degree stays $O_{\theta, \varepsilon}(1)$ throughout the $O_{\theta, \varepsilon}(1)$ recursion stages, so $|E| = O_{\theta, \varepsilon}(1)$. Outside E , this normalization does not change the specialized solution set.

A terminal recursion polynomial is independent of all derivative variables. Outside E it is a nonzero polynomial in X , so it cannot vanish identically in X . Thus no polynomial solution survives past the terminal stage. Summing the regular branches proves the proposition. $\square$

5 From Polynomial Solution Families to MCA

Fix $u_0, u_1 \in \mathbb{F}_q^n$. Fix one algebraic set B from Proposition 4.2 and its associated polynomial P . For every $i \in [n]$, define

$$F_i = P(\alpha_i) - u_{0,i} - zu_{1,i}. \tag{13}$$

By Proposition 4.2,

$$\deg F_i = O_{\theta, \varepsilon}(n^2). \quad (14)$$

For every irreducible algebraic subset $V \subseteq B$, let $\text{Bad}(V)$ denote the bad parameters represented by points of V .

We first handle the case where many agreement equations vanish on all of V .

Lemma 5.1. *If*

$$V \subseteq \mathcal{Z}(F_{i_1}, \dots, F_{i_k})$$

for distinct $i_1, \dots, i_k$, then

$$|\text{Bad}(V)| \leq n.$$

Proof. Interpolate the values u_{0,i_j} and u_{1,i_j} on the k distinct points α_{i_j} . This gives unique polynomials $P_0, P_1 \in \mathbb{F}_q[X]_{<k}$. On V , the polynomial P agrees with $P_0 + zP_1$ at those k points, so

$$P = P_0 + zP_1$$

identically on V .

Let

$$T = \{i : P_0(\alpha_i) = u_{0,i} \text{ and } P_1(\alpha_i) = u_{1,i}\}.$$

For $i \notin T$, the equation

$$P_0(\alpha_i) - u_{0,i} + z(P_1(\alpha_i) - u_{1,i}) = 0$$

is a nonzero affine equation in z , so it holds for at most one value of z . A bad witness must use at least one coordinate outside T ; otherwise (P_0, P_1) would explain it. Assign each bad parameter to one such coordinate. Two different bad parameters cannot be assigned to the same coordinate, so there are at most n of them. $\square$

We now count all remaining bad parameters. Notice that

$$A - k + 1 \geq \theta \varepsilon n - O(1). \quad (15)$$

Lemma 5.2. *There is $C_{\theta, \varepsilon} > 0$ such that every irreducible algebraic subset $V \subseteq B$ satisfies*

$$|\text{Bad}(V)| \leq \deg(V) n (C_{\theta, \varepsilon} n^2)^{\dim V}.$$

Proof. We induct on $\dim V$. The case $\dim V = 0$ is immediate because a zero-dimensional variety has at most $\deg V$ points.

If at least k agreement polynomials vanish identically on V , Lemma 5.1 gives $|\text{Bad}(V)| \leq n$, which is already enough. So assume at most $k - 1$ of them vanish identically.

Choose one representing point of V for each parameter in $\text{Bad}(V)$. These points have distinct z -coordinates. Each point satisfies at least A agreement equations, and at most $k - 1$ of those equations contain all of V . Hence each point lies on at least $A - k + 1$ equations that properly cut V . Double counting these incidences gives

$$(A - k + 1) |\text{Bad}(V)| \leq \sum_{i: V \not\subseteq \mathcal{Z}(F_i)} \sum_{W \in \text{Irr}(V \cap \mathcal{Z}(F_i))} |\text{Bad}(W)|. \quad (16)$$

Overcounting on the right is harmless.

By (14) and Bézout,

$$\sum_{W \in \text{Irr}(V \cap \mathcal{Z}(F_i))} \deg W \leq C_{\theta, \varepsilon} \deg(V) n^2.$$

Every such W has dimension at most $\dim V - 1$, so the induction hypothesis gives

$$\sum_{W \in \text{Irr}(V \cap \mathcal{Z}(F_i))} |\text{Bad}(W)| \leq C_{\theta, \varepsilon} \deg(V) n^3 (C_{\theta, \varepsilon} n^2)^{\dim V - 1}$$

after enlarging the constant. There are at most n choices of i , while (15) gives $A - k + 1 \geq c_{\theta, \varepsilon} n$ for large enough n . Substituting into (16) and dividing by $A - k + 1$ yields the claimed bound, after enlarging $C_{\theta, \varepsilon}$ once more. $\square$

We can now prove the main result.

Theorem 5.3 (Near-Capacity Mutual Correlated Agreement). *For every $u_0, u_1 \in \mathbb{F}_q^n$,*

$$|\text{Bad}_A(u_0, u_1)| \leq n^{C_{\theta, \varepsilon}}$$

for some $C_{\theta, \varepsilon} > 0$. Consequently,

$$\varepsilon_{\text{MCA}}(\text{RS}_{<k}(\alpha_1, \dots, \alpha_n), A) \leq \min \left\{ 1, \frac{n^{C_{\theta, \varepsilon}}}{q} \right\}.$$

Proof. For each bad parameter outside the exceptional set E from Proposition 4.2, choose one witness polynomial and one point on a solution-family component that represents it. Assign that point to one irreducible component V containing it.

Lemma 5.2 bounds the number assigned to V by

$$\deg(V) n (C_{\theta, \varepsilon} n^2)^{\dim V}.$$

Proposition 4.2 gives $\dim V \leq d + 2 = O_{\theta, \varepsilon}(1)$ and says that the sum of the degrees of all components is $O_{\theta, \varepsilon}(n^2)$. Summing over V therefore gives only $n^{O_{\theta, \varepsilon}(1)}$ bad parameters outside E . Since $|E| = O_{\theta, \varepsilon}(1)$, the same bound holds for all bad parameters. Dividing by q proves the MCA error bound. $\square$

For any other fixed rate $R < \varepsilon$, the same proof applies with slack $1 - R/\varepsilon$ and the corresponding interpolation parameters from [BCPZZ26]. Unlike ordinary list decoding, this does not follow just by passing to a subcode.

6 Discussion and Open Problems

Theorem 5.3 shows that the near-capacity list-decoding radius from Theorem 4.1 of [BCPZZ26] survives when we move from one received word to an affine line. The interpolation step is almost entirely inherited from that work. What is new here is the simultaneous control of all polynomial solutions as the line parameter varies.

Our bound on the number of bad parameters is polynomial rather than linear in n . The loss comes from the degrees of the solution families and agreement equations, which are paid through Bézout each time the dimension drops. At the Johnson radius, factorization arguments give a linear total useful-factor degree and hence $O(n/q)$ MCA error [Hab25, Ben+25]. This leaves a natural question.

Open Problem 6.1. *Can one prove*

$$|\text{Bad}_A(u_0, u_1)| = O_{\theta, \varepsilon}(n)$$

for the parameters of Theorem 5.3?

A positive answer would give nontrivial MCA over fields of size $q = \Theta_{\theta, \varepsilon}(n)$ at the same near-capacity radius as the list decoder.

It would also be interesting to go beyond affine lines. The interpolation step already extends to polynomial families of received words by replacing $u_0 + Zu_1$ with a low-degree polynomial in Z . The remaining challenge is to control the resulting polynomial solution families and agreement equations.

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